

Task 3: Command-Line Administration Guide

Introduction

This document demonstrates basic Linux command-line administration commands used for file management, user permissions, process monitoring, and software installation. These commands are important for system administrators and IT support technicians when managing Linux systems.

1. pwd Command

Purpose

The pwd command stands for “Print Working Directory.” It displays the current directory the user is working in.

Command

```
pwd
```

Example Output

```
/home/student/Documents
```

Explanation

This command helps users identify their current location within the Linux file system.

2. ls Command

Purpose

The ls command lists files and directories in the current folder.

Command

```
ls
```

Additional Options

```
ls -l
```

Explanation of -l

Displays files in a detailed format including:

- Permissions
- Owner

- File size
- Date modified

Example Output

drwxr-xr-x 2 user: user 4096 May 29 Projects

-rw-r--r-- 1 user: user 120 May 29 notes.txt

3. cd Command

Purpose

The cd command stands for “Change Directory.” It is used to move between folders.

Commands and Examples

Move into a folder

cd Documents

Move back one directory

cd ..

Go to home directory

cd ~

Explanation

This command allows navigation throughout the Linux file system.

4. mkdir Command

Purpose

The mkdir command creates a new directory (folder).

Command

mkdir Projects

Example

mkdir LinuxFiles

Explanation

This command is used to organize files into folders and create project directories.

5. touch Command

Purpose

The touch command creates a new empty file.

Command

```
touch notes.txt
```

Example

```
touch report.txt
```

Explanation

This command is commonly used to quickly create files before editing them.

6. chmod Command

Purpose

The chmod command changes file permissions.

Command

```
chmod 755 script.sh
```

Explanation of Permission 755

- Owner: Read, Write, Execute
- Group: Read, Execute
- Others: Read, Execute

Another Example

```
chmod 644 notes.txt
```

Meaning of 644

- Owner: Read and Write
- Group: Read only
- Others: Read only

Explanation

This command helps protect files and controls who can access or modify them.

7. chown Command

Purpose

The chown command changes file ownership.

Command

```
sudo chown user notes.txt
```

Example

```
sudo chown elizabeth project.txt
```

Change Owner and Group

```
sudo chown elizabeth:developers project.txt
```

Explanation

This command is useful in multi-user systems where ownership and access control are important.

8. ps Command

Purpose

The ps command displays currently running processes.

Command

```
ps
```

Common Option

```
ps aux
```

Example Output

```
USER      PID %CPU %MEM    COMMAND
root         1  0.0  0.1   systemd
user    2045  1.2  2.5    Firefox
```

Explanation

This command helps administrators monitor active applications and troubleshoot problems.

9. top Command

Purpose

The top command provides real-time system monitoring information.

Command

top

Information Displayed

- CPU usage
- Memory usage
- Running processes
- System performance

Exit Command

Press:

q

Explanation

This command helps identify applications using excessive system resources.

10. apt install Command

Purpose

The apt install command installs software packages on Debian-based Linux systems such as Ubuntu.

Command

sudo apt install git

Update Package List First

sudo apt update

Install Multiple Packages

sudo apt install git curl vim

Explanation

This command downloads and installs software from online repositories.

Practical Demonstration Example

The following example demonstrates how multiple commands can be used together:

```
pwd
```

```
mkdir LinuxPractice
```

```
cd LinuxPractice
```

```
touch notes.txt
```

```
ls -l
```

```
chmod 644 notes.txt
```

Explanation of Steps

1. Displays the current directory.
2. Creates a new folder called LinuxPractice.
3. Opens the folder.
4. Creates a file named notes.txt.
5. Displays the file details.
6. Changes file permissions.

Conclusion

Linux command-line administration commands are essential for managing files, directories, users, permissions, processes, and software installations. Understanding these commands improves system administration skills and allows efficient management of Linux operating systems.